

Cells Test - Answer Key

1. Describe the main stages of cellular respiration and their locations within the cell.

Answer: Cellular respiration consists of three main stages: glycolysis, the Krebs cycle, and the electron transport chain. Glycolysis occurs in the cytoplasm and produces a small amount of ATP. The Krebs cycle occurs in the mitochondrial matrix, and the electron transport chain occurs in the inner mitochondrial membrane. The electron transport chain produces the majority of the cell's ATP through a process called chemiosmosis.

Explanation: Cellular respiration is the process by which cells convert nutrients into usable energy in the form of ATP.

2. Compare and contrast prokaryotic and eukaryotic cells, including their structures and functions.

Answer: Prokaryotic cells lack a membrane-bound nucleus and most organelles, whereas eukaryotic cells contain a true nucleus and a variety of specialized organelles. Prokaryotic cells are typically found in bacteria and archaea, while eukaryotic cells are found in plants, animals, fungi, and protists. The presence of a true nucleus and organelles in eukaryotic cells allows for greater complexity and specialization of cellular functions.

Explanation: The cell theory states that all living things are made of cells, and there are two broad categories of cells: prokaryotic and eukaryotic.